

Project: Causal Machine Learning

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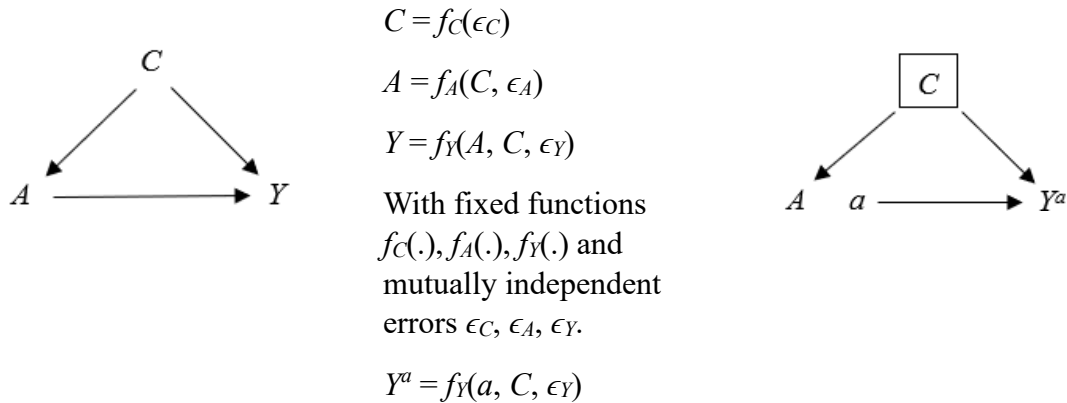
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Question 1

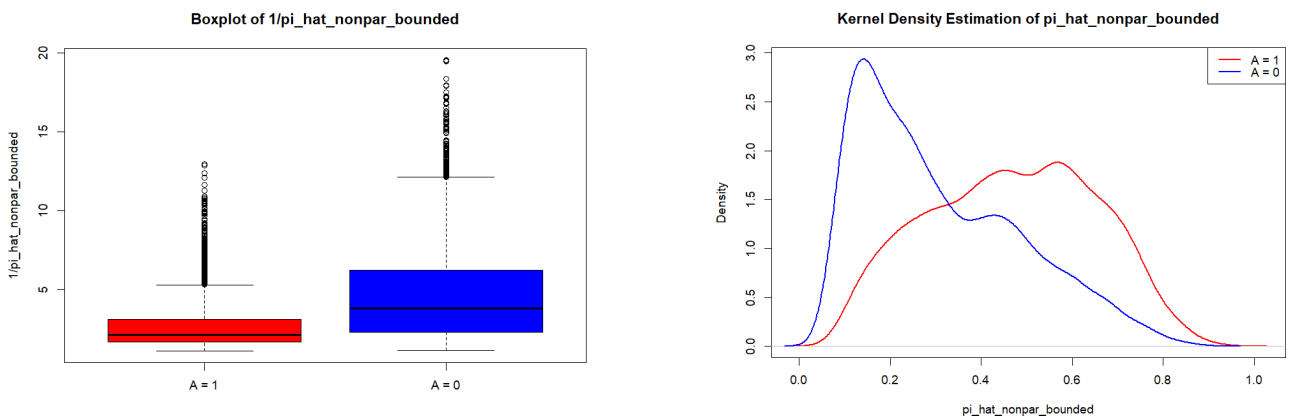
What needs to be verified here is $Y^a \perp\!\!\!\perp A \mid C$ for all a (conditional exchangeability); with Y = accumulated assets, A = eligibility for enrolling in a 401(k) plan, C = income. This can be done through constructing the following causal Directed Acyclic Graph (DAG) representing the current situation, to then specify which manner of 'data generation' this causal DAG suggests and to formalize it as a nonparametric structural equation model (a system of equations), to then construct a Single World Intervention Graph (SWIG) based on this system of equations. This SWIG is crucial to the verification process as it includes the necessary counterfactual.



Based on this SWIG, the abovementioned conditional exchangeability assumption holds when conditioning on C . Specifically, following the rules of d-separation, conditioning on C closes the otherwise open backdoor path between A and Y^a by turning this non-collider into a collider on this path. Therefore, the association between A and Y in the earlier causal DAG is unconfounded after controlling for C and the causal effect of A on Y can be estimated (assuming C contains all confounders).

Question 2

To calculate both the AIPW and TMLE estimator we first investigated the positivity assumption. This was done by looking at the extent of overlap in the propensity scores for the eligible and ineligible groups. The overlap was limited, which could cause a highly variable ATE estimate with high standard errors. Based on this the average treatment effect on the treated (ATT) seemed most suitable.



To calculate the AIPW based ATT the npcausal package was used. The ATT $E[Y^1 - Y^0 \mid A = 1]$ estimate was 9458.67 (95% CI: 6491.35 - 12426.00; $p < 0.001$). This implies that within the

subgroup of the people that are eligible, net financial assets are on average about \$9459 higher than they would have been without eligibility. When running the analyses without covariate adjustments, the estimate is 19555.51, (95% CI: 17010.64 - 22100.18). This is around a two-fold increase. Indicating that failing to adjust for the confounders leads to an upward bias. By including covariates, the adjusted AIPW ATT the effect of eligibility on net financial assets itself.

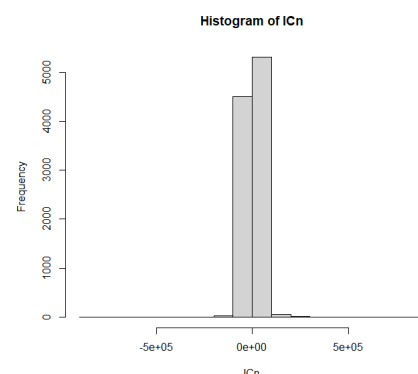
For the Targeted Maximum Likelihood estimator (TMLE) based ATT again five-fold cross-fitting was used to first estimate the nuisance parameters. This was done with a SuperLearner library of "SL.glm", "SL.step", "SL.glm.interaction", "SL.ranger". For each fold we predicted the control-outcome regression Q_0 and the treatment propensity g , bounded g between 0.01 and 0.99, and constructed the clever covariate H . Then, a Gaussian fluctuation model updated Q_0 to Q_0^* , and the ATT was computed as the average of $Y - Q_0^*$ among the treated. The obtained estimate was 5806.98, (95% CI: 1 336.41 - 10 277.55). For among those eligible for a 401(k), TMLE estimates that eligibility raises net financial assets by about \$5807 on average. When running the analyses without covariate adjustments, the estimate is 7266.49, (95% CI: 4489.73 – 10043.26). So, on average, 401(k)-eligible households have about \$7266 more in net financial assets than ineligible ones when you don't adjust for any covariates. Again, showing an upward bias introduced by the confounders.

Overall, both the AIPW and TMLE show a clear positive effect of 401(K) eligibility on net financial assets. Additionally, it also shows the importance of including the covariates for credible causal inference

Question 3

The performed analyses are as follows. First, A (eligibility for enrolling in a 401(k) plan) was predicted based on L (see question 2) by training a SuperLearner model (family: binomial) on A and L using a library of GLM, Step, and Ranger algorithms. Second, Y (net financial assets) was predicted based on L by training a SuperLearner model (family: gaussian) on Y and L using a library of GLM, Step, and Ranger algorithms. Third, the influence function values for all individuals in the sample were calculated. Importantly, 5-fold cross-fitting was applied for all previous analyses. Fourth, the estimate for θ was obtained by taking the sample average of all previous influence function values. Fifth, standard errors and 95% confidence intervals were calculated. Finally, a one-sample t-test was performed of the null hypothesis that θ equals 0. The table below shows relevant results. See our R script for further details on the conducted analyses. The estimate of θ is an estimate of the expected (population average) covariance between eligibility for enrolling in a 401(k) plan and the net financial assets, conditional on the control variables mentioned in question 2. This expected covariance is significantly (with 0.05 significance threshold) different from 0 and seems to be positive, indicating a positive association between eligibility for enrolling in a 401(k) plan and net financial assets (which can be interpreted as a causal effect of enrolling on net financial assets, assuming all relevant control variables are included). Quality check using a histogram (see below) of all individuals' influence function values (ICn) did not result in extreme or unexpected values being detected visually.

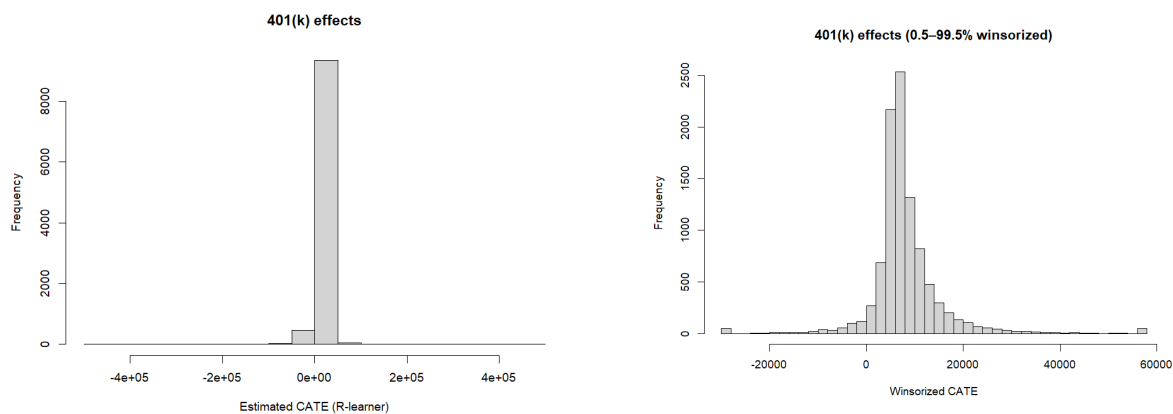
```
A tibble: 1 x 8
  Method Estimate StdError CI_Lower CI_Upper T_value P_value Df
<chr>      <dbl>    <dbl>   <dbl>   <dbl>   <dbl>   <dbl>  <dbl>
Eff. Influence Function 2556.    320.    1930.    3183.    7.99 1.45e-15  9914
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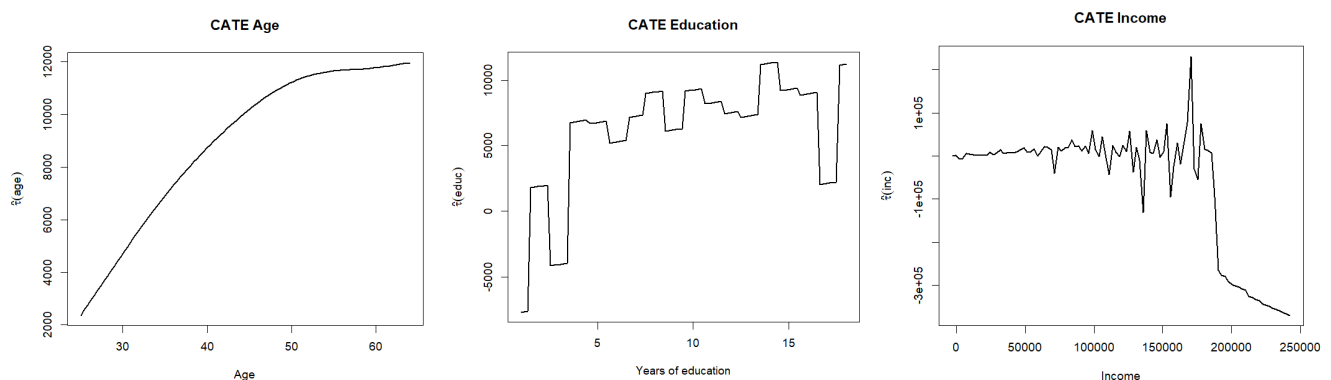
Question 4

To estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets in function of age, income, and years of education, while controlling for covariates, we applied the R-learner with five-fold cross fitting. The SuperLearner libraries used were “SL.glm”, “SL.ranger, and “SL.glmnet”. For the general structure of the analyses: a prediction model was trained for the outcome, including all the covariates. Then the propensity scores were estimated. These predictions were then used to construct the pseudo-outcome and weights. Finally, predictions were made in function of age, income and years. See our R-script, for more details on the analyses.

To investigate the distribution of the effect estimates for all individuals two histograms were plotted. The first plot shows the raw distribution of the CATES with a clear spike clustered very close to zero and a few extreme outliers. To further investigate the core of the first plot we have winsorized the tails (0.5-99.5%) to create the second plot. This allows to see the heterogeneity of the effect between about -5000 and 20000, with a mode around 7600. Showing that most individuals benefit positively, but that there also some who experience a minimal or even negative effect, and some gain substantially more than the average. The estimated effect for this analyses is 7626.35 (95% CI: 7054.62, 8198.08), which closely matches the mode of the winsorized distribution.



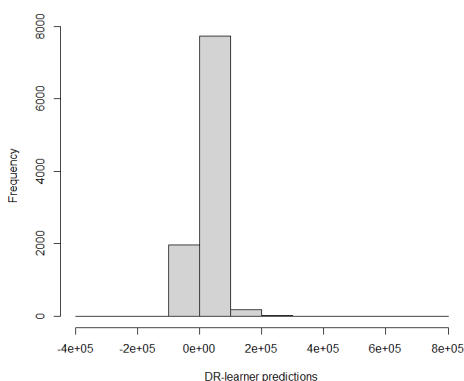
To investigate how the estimated conditional average treatment effects vary by age, income and years of education, three plots were constructed. For the age plot, there is a consistent increase, where older individuals experience larger net financial asset gains from eligibility. The education-CATE plot shows a stepwise increase, indicating that individuals with longer education experience greater benefits, starting from slightly negative effects. For the income



CATE plot, the effect remains stable for the low and middle-income levels, followed by a spike near \$175000 annual income, and a sharp decline for the highest earners.

Question 5

The performed analyses are as follows. First, A (eligibility for enrolling in a 401(k) plan) was predicted based on L (see question 2) by training a SuperLearner model (family: binomial) on A and L using a library of GLM, Step, and Ranger algorithms. Second, Y (net financial assets) was predicted based on L by training a SuperLearner model (family: gaussian) on Y and L in the treated subgroup using a library of GLM, Step, and Ranger algorithms. Third, the pseudo-outcomes C for all individuals in the sample were calculated. Fourth, these pseudo-outcomes were predicted based on age, income, and years of education by training a SuperLearner model (family: gaussian) on these pseudo-outcomes and age, income, and years of education using a library of GLM, Step, and Ranger algorithms. 5-fold cross-fitting was applied in all previous analyses. The histogram below illustrates these final predictions. It appears that the counterfactual net financial assets that would be observed when being eligible for enrolling in a 401(k) plan in function of age, income, and years of education (while controlling for the variables in question 2) is for most people (in the sample) positive, but still for a substantial portion of people is this amount negative. The fifth and sixth analyses concerned again predicting A based on L and predicting Y based on L in the treated subgroup following the aforementioned procedures. The seventh analysis concerned predicting the earlier obtained DR-learner predictions based on L (see question 2) using the same SuperLearner model configurations as in previous analyses. Lastly, influence function values (this efficient influence function is based on the one for ATE but slightly adjusted, see R script for details) were calculated for all individuals in the sample and its sample average was taken to determine the estimate of the counterfactual prediction error. 5-fold cross-fitting was again applied in previous relevant analyses. This estimate appears to be incredibly large. Corresponding 95% confidence interval was calculated. The table below shows relevant results.



```
A tibble: 1 × 5
  Method      Estimate StdError CI_Lower CI_Upper
<chr>      <num:.3!> <num:.3!> <num:.3!> <num:.3!>
Eff. Influence Function 2.406e10  3.350e9  1.749e10  3.062e10
```

Acknowledgement Individual Contribution:

- Florin Moldovan: Exercises 1 and 3
- Thomas De Schryver: Exercises 2 and 4
- Both authors: Jointly completed Exercise 5 and reviewed each other's work.